Ana III Zusammenfassung

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1 Measure Theory

Definition 1.1 (outer measure)**.** Let X be any set and let $\mu : 2^X \to [0, \infty]$ then μ is an outer measure if

1. $\mu(\emptyset) = 0$
2. $\mu(A) \leq \sum_k \mu(A_k)$ for every $A \subset \bigcup_k A_k$

Definition 1.2 (measurability)**.** A set $A \subset X$ is considered μ-measurable if for every $B \subset X$ we have the inequality

$$\mu(B) \geq \mu(A \cap B) + \mu(B \backslash A)$$

Now let $\mathcal{M}(\mu)$ be the set of all μ-measurable sets.

Theorem 1.3 (basic properties of measurable sets)**.** *Let $A \subset X$ be measurable and let $A_k \subset X$ be a sequence of measurable sets then we have the following properties*

1. $X \setminus A = A^C \in \mathcal{M}(\mu)$
2. $\bigcap_k A_k \in \mathcal{M}(\mu)$ *and* $\bigcup_k A_k \in \mathcal{M}(\mu)$
3. *if A_k is a pairwise disjoint sequence then we have* $\mu(\bigcup_k A_k) = \sum_k \mu(A_k)$

Proof. will be added later □

Definition 1.4 (measure space)**.** A triple $(X, \mathcal{M}, \mu)$ is called a measure space if $\mu : \mathcal{M} \to [0, \infty]$ is a measure and $\mathcal{M}$ is a σ-Algebra.

Theorem 1.5 (monotone convergence for measures)**.** *Let $(A_n)_n$ be a sequence of measurable sets*

1. *if $A_1 \subset A_2 \subset \dots$ then we have*

$$\mu(\bigcup_n A_n) = \lim_{n \to \infty} \mu(A_n)$$

2. *if $A_1 \supset A_2 \supset \dots$ and $\mu(A_1) < \infty$ then we have*

$$\mu(\bigcap_n A_n) = \lim_{n \to \infty} \mu(A_n)$$

Proof. will be added later □

Definition 1.6 (σ-finite)**.** A measure space $(X, \mathcal{M}, \mu)$ is σ-finite if there are sets $E_n \in \mathcal{M}$ such that

$$X = \bigcup_n E_n, \quad \mu(E_n) < \infty$$

1.1 Lebesgue Measure

Definition 1.7 (rectangle)**.** Let $a, b \in \mathbb{R}^n$ such that $a_i \leq b_i$ then we define the rectangle with corners a and b by

$$Q(a,b) := (a_1, b_1) \times \cdots \times (a_n, b_n)$$

and we define the volume with

$$|Q(a,b)| := \prod_{i=1}^{n} (b_i - a_i)$$

Definition 1.8 (lebesgue outer measure)**.** Let $E \subset \mathbb{R}^n$ then we define the n dimensional Lebesgue Outer Measure by

$$\mathcal{L}^n(E) := \inf\{\sum_k |Q_k| : E \subset \bigcup_k Q_k, \text{ where } Q_k \text{ are rectangles}\}$$

Theorem 1.9 (properties of the lebesgue measure)**.** *The Lebesgue measure has the following properties*

1. $\mathcal{L}^n$ *is Borel regular and* σ*-finite*
2. $\mathcal{L}^n(Q(0,1)) = 1$
3. $\mathcal{L}^n(A + t) = \mathcal{L}^n(A)$ *where* $t \in \mathbb{R}^n$
4. $\mathcal{L}^n(\Lambda(A)) = \det(\Lambda)\mathcal{L}^n(A)$ *with* $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ *and* $\lambda_k \geq 0$ *for all* k

Proof. will be added later □